

Three-Axial Helmholtz Coil Design and Validation for Aerospace Applications

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This paper presents the detailed design, construction, and validation of a three-axis square Helmholtz coil. It also describes the methodology used to drive each pair of coils as well as the setup to operate it in a closed-loop system using a digital PID controller. The coil will be mainly used for aerospace applications, especially to aid the development and testing of attitude determination and control systems that use the earth's magnetic field as a reference vector. Most of the system was built using commercial components, reducing cost, and complexity compared to similar commercial systems. The assembled Helmholtz coil has approximately one cubic meter and can generate magnetic fields up to 2 G/axis, keeping a uniformity of 0.04% around 11 cm of the center, in each axis. A custom-designed voltage-controlled current source, based on the Howland current pump, was employed, requiring no complex electronic circuits. The coil was designed to be part of a hardware-in-the-loop (HiL) system, which is controlled by a dSPACE modular simulation hardware and uses a commercial fluxgate magnetometer as the reference. This setup reduces the complexity of the proposed system when compared to similar ones. This paper presents two distinct results: first, there is the validation and results of the uniformity regarding the generated field around the system's center; second, results of the setup with the closed-loop HiL simulation are shown, which includes tests of the coil when generating a dynamic magnetic field.

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I. INTRODUCTION

The development of magnetic field simulators has been a subject of study for many decades. Several areas of study use magnetic fields for different purposes, from biomedical to aerospace engineering systems. In terms of the latter, geomagnetic field measurements contain information used by attitude determination systems for airplanes and satellites [1], for example.

The most commonly implemented magnetic field simulators found in the literature are the Helmholtz, Merritt, and Ruben coil systems, which are all discussed in [2]. The Helmholtz coil is the simplest and used in several works because it has only two windings per axis. It basically consists of two parallel coaxial coils. Due to the symmetry of its structure and the proximity between both coils, it is possible to generate a uniform magnetic field in one direction on a certain region around the center of the set. In order to generate a field in any direction, one uses three orthogonal Helmholtz coils aligned to each of the x -, y -, z -axes of a rectangular coordinate system.

Though the theory behind the Helmholtz coil is relatively well known, the design and construction of a triaxial coil is not trivial and commercial devices are quite expensive. Moreover, the specifications of a magnetic field simulator, such as the total field intensity and its uniformity within a volume, the transient response time, the coil's inductance, among others, can vary significantly depending on the application. For instance, [3] shows a system capable of generating magnetic fields greater than 20 G in a small volume for bioelectromagnetics application, while [1] describes a very complex system that can produce a homogeneous field in a large volume for nanosatellite validation.

This paper presents the design, implementation, and validation of a triaxial Helmholtz coil of which the main applications will be the validation of attitude determination systems and the calibration and testing of low-cost magnetic sensors. Two recent examples of design and manufacturing of triaxial systems for similar applications are given in [4] and [5]. Both were developed for geomagnetic field simulation for earth orbit. In turn, the works in [6]–[8] show similar uses of the Helmholtz system but do not present a detailed discussion about the coils itself.

The detailed requirements of the proposed system and how they were defined are discussed in Section II. At this point, it is emphasized that the uniformity of the field within the desired volume inside the system is a crucial design parameter. Sections III-B and III-C discuss the square Helmholtz coil's uniformity and further information is found in [9]–[11]. Also, as discussed in [12], the frequency response to variable currents is essential if the goal is to simulate aerospace applications in real time, which is also the aim of this paper. Tests under these conditions are presented.

There is also a discussion on the current source used to drive the Helmholtz coil as well as the setup to operate it with a digital closed-loop control system. Most works

in the literature do not show how the current drive is implemented and controlled. The proposed voltage-controlled current driver is shown in Section VII-A and consists of a Howland current pump (HCP) [13]. The electronic system employed is very simple and capable of driving the current using linear circuits. Therefore, disadvantages typical of nonlinear or *H bridge* drivers, such as the zero-crossing problem, are avoided. Also, implementing a closed-loop system becomes much simpler.

With respect to the closed-loop control, it is usually performed by means of an electronic system consisting of a PID controller. Although the theory is simple, its implementation and tuning might prove challenging, as in the prototype presented in [4]. To overcome this problem, this paper uses a hardware-in-the-loop simulation based on dSPACE modular hardware. This system is fully compatible with MATLAB/Simulink model design, which greatly simplifies electronic circuits, thus reducing time needed for testing and validation. Basically, the only electronics needed is the Howland current driver that is controlled by the digital-to-analog converter on the dSPACE hardware. This setup allows fast evaluation and modifications on the Helmholtz coil controller and the complete integration with Simulink software develop to test both nanosatellite attitude control and magnetic sensors. Furthermore, it is possible to implement the control system as a low-cost embedded electronic system, using the automatic code generation available in the Simulink/dSPACE tools.

A. Contributions of This Paper

The original contribution of this paper is the unveiling of the design details of a low cost three-dimensional (3-D) Helmholtz coil that has enough performance to be applied as a tool for aerospace applications, specifically for the development and testing of nanosatellite attitude determination and control systems based on low-cost magnetic sensors.

To accomplish this, the paper shows a simple and inexpensive construction methodology and materials to build the windings and to assemble them as a 3-D Helmholtz coil, following its design calculations. This paper also reports the successful use of a much simpler electronic circuit to drive the current through the coils, when compared to the solutions found elsewhere.

The topology used is the modified HCP, which retains a linear response during zero crossing of the current signal. This paper presents a control system for the magnetic field produced by the coil based on the PID strategy. It was implemented in the digital domain using a Simulink-dSPACE dedicated system, which enormously simplified its development. This methodology is readily suitable to be applied in the development of a low cost embedded control system.

At last, a dynamic analysis is presented by submitting the Helmholtz coil to both step and sinusoidal inputs, considering the system in the closed-loop setup. Such tests are important to evaluate the system response and whether it meets the requirements for the desired applications.

B. How this paper is organized

Section II presents the main aspects and requirements of the project to meet the needs of the proposed applications, followed in Section III by the theoretical study and basis for a square Helmholtz coil system. Section IV brings a very comprehensive description of the structural and electrical design chosen for the constructed coil as well as the prototype results. The remaining sections present the results, which are divided in two parts. First, the uniformity was tested by a specific setup and then the final prototype system was tested with dSPACE hardware running the HiL simulation.

II. COIL DESIGN MAIN REQUIREMENTS

A decision was made to build a Helmholtz coil with square shape and rounded corners, rather than a circular one, due to the simplicity of the construction and assembly procedure. Although circular coils simplify the winding process, its manufacturing and fastening structures between pairs are far more complex, resulting in higher cost and less design flexibility [9]. It was also implied that the square structure could be simply built using straight commercial aluminum profiles, that are commonly used as structural elements for various applications. This concept greatly reduces the effort and cost for building a relatively large structure and also adds flexibility to the system. In addition to the mechanical advantages, the square system also produces a slightly better uniformity region than a circular coil of same size [11].

Other important requirements are the maximum field strength and the uniformity percentage needed. Since the applications mainly consist of the validation and testing of navigation equipment that uses magnetic sensors and the geomagnetic field simulation for aerospace missions, the field generation capacity was decided to be at least enough to compensate the earth's field (creating a field in the opposite direction in order to *cancel* it) and subsequent generation of a new field of strength up to 1.2 G in either direction (coordinate). Thus, it was established that each pair of coils must generate a field strength of 2.0 G.

The uniformity region must be large enough to allocate the magnetic sensor that will be used as reference for the closed-loop control and further equipment or sensors that will be evaluated. The low-cost attitude determination system described in [14], which seeks to use low-cost sensors for nanosatellite applications, places its sensors in a volume of approximate 8 cm × 8 cm × 8 cm. An 11 cm-sized cube is necessary to place this equipment and the fluxgate magnetometer used as reference inside the coil. The magnetic field must be uniform enough so that the resolution of the sensors under test are not capable of detecting such variation, regardless of the position within the delimited volume. Assuming a full scale of 1.2 G and 0.02% uniformity per axis, this means 240 μ G of variation in each axis. Such value is sufficiently small not to be detected by most of the low-cost magnetic sensors.

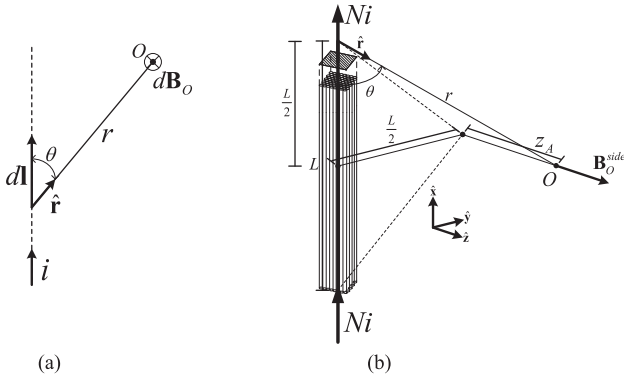


Fig. 1. (a) Biot-Savart Law: Magnetic field produced at the point O given a current i in a differential length dl at a distance r . (b) Field contribution generated in O by only one side of the coil.

At last, the coil update rate and its frequency response should be as high as possible. To achieve that, it is important to keep the coil inductance as small as possible, in order to reduce the time constant of the LR circuit and, thus, improve its frequency response.

III. THEORETICAL BASIS FOR SQUARE COILS

In order to design a Helmholtz coil system, it is necessary to predict the behavior of the generated magnetic field in the neighborhood of the coil's center. To simplify the analysis, the magnetic field equation is deduced based on the current, number of turns, size of the coils, and considering the optimum distance between pairs. The uniform region that meets the design criteria is explained below. With these two theoretical concepts, as well as the requirements discussed, it is possible to specify the rest of the system. Part of the following mathematical analysis is based on [9].

A. Mathematical Modeling

The magnetic field $\mathbf{B}$ at a given point O due to a current element $d\mathbf{l}$ [see Fig. 1(a)] is obtained by the well known Biot-Savart law, which describes the magnetic field generated by an electric current i . This is described as

$$d\mathbf{B}_O = \frac{\mu_0 i}{4\pi r^2} d\mathbf{l} \times \hat{\mathbf{r}}. \quad (1)$$

The field is proportional to i , dl , and $\sin \theta$, and inversely proportional to r^2 , where θ is the angle between $d\mathbf{l}$ and $\hat{\mathbf{r}}$. Its direction and sign is given by the cross product rule. Finally, μ_0 is the magnetic permeability of vacuum.

However, an electric current is always associated with a closed circuit so there is no reason to break it into multiple elements, except as an auxiliary step in the calculation. Therefore, it is necessary to integrate the current elements along each side of the square coil.

Considering a group of straight wires in one side of the coil [see Fig. 1(b)] of length L , all current elements contribute $d\mathbf{B}_O$ to O with the same sign and direction. Hence, it suffices to integrate $d\mathbf{B}_O$ from $-L/2$ to $+L/2$. The magnitude of the produced field in O by the flow of

current in one side of the coil is given as

$$B_O^{\text{side}}(z_A) = \frac{\mu_0 N i}{4\pi} \frac{L}{2} \int_{-L/2}^{L/2} \frac{dx}{\left(z_A^2 + \left(\frac{L}{2}\right)^2 + x^2\right)^{3/2}} \quad (2)$$

where N is the number of turns, i is the current in each wire, and z_A is the distance from point O to the center of the coil. Also, the following equations must be considered as:

$$\sin \gamma = \frac{L/2}{p}; \quad \sin \theta = \frac{p}{r} \quad (3)$$

$$p = \sqrt{(L/2)^2 + z_A^2}; \quad r = \sqrt{x^2 + z_A^2 + (L/2)^2}. \quad (4)$$

Using the identity (5) on (2), the contribution of one side of the coil to the generated field at a distance z_A can be calculated. This contribution is given by (6)

$$\int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}} \quad (5)$$

$$B_O^{\text{side}}(z_A) = \frac{\mu_0 N i}{\pi} \frac{L^2}{(4z_A^2 + L^2) \sqrt{4z_A^2 + 2L^2}}. \quad (6)$$

Owing to symmetry, the intensity of the field generated by a single pair of Helmholtz coils at a distance z_A can be calculated by summing the contribution of each of the four sides. Fig. 2 represents this idea while the following equations show the total intensity:

$$B_O^{\text{total}}(z_A) = 4B_A^{\text{side}}(z_A) + 4B_A^{\text{side}}(-z_A) \quad (7)$$

$$B_O^{\text{total}}(z_A) = \frac{8\mu_0 N i L^2}{\pi (4z_A^2 + L^2) \sqrt{4z_A^2 + 2L^2}}. \quad (8)$$

Assuming that $z_A = \sigma L/2$, where σ is a parameter that maximizes the uniformity of the field around the coil center, the expression of the field produced by the Helmholtz coils at the origin O related to this parameter is given as

$$\mathbf{B}_O^{\text{total}}(\sigma) = \frac{8\mu_0 N i}{\pi L} \frac{1}{(\sigma^2 + 1) \sqrt{\sigma^2 + 2}} \hat{\mathbf{z}} \quad (9)$$

which defines the field intensity on the center point as a function of σ . This parameter is a constant that relates the distance between coils to the length L . Section III-B demonstrates that there is a distance that maximizes the field uniformity around the coil center. With that distance, it is possible to define the magnetic field depending only on i , L , and N .

B. Optimal Distance Between the Coils

As discussed both in [9] and [10], the maximization of the field uniformity around the origin point O is related to the distance (σL) between the coils. That is, there is an optimum distance where the variation of the field is minimized along the symmetry axis and around that point.

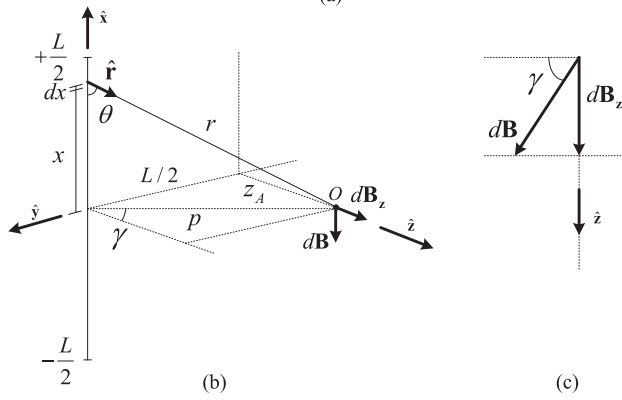
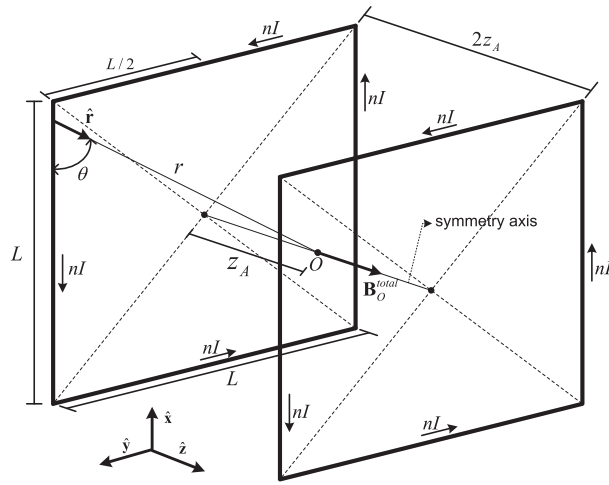


Fig. 2. Perspective of the Helmholtz coil and consequent representation of the magnetic field generated at the origin of the system.

The aim here is to reduce the field variation, which is given by the first derivative of (9). Therefore, the field variation is described as

$$\frac{dB_O^{\text{total}}}{d\sigma} = \frac{8\mu_0 Ni}{\pi L} \left(\frac{-\sigma(3\sigma^2 + 5)}{(\sigma^2 + 1)^2(\sigma^2 + 2)^{3/2}} \right). \quad (10)$$

As a function assumes its minimum when its first derivative is null, it is necessary to find a value of σ that nulls the second derivative of (9), which is shown in (11). This condition is satisfied at the roots of the numerator polynomial of the following equation, which is true for $\sigma = \pm 0.544505643$:

$$\frac{d^2 B_O^{\text{total}}}{d\sigma^2} = \frac{8\mu_0 Ni}{\pi L} \left(\frac{2(6\sigma^6 + 18\sigma^4 + 11\sigma^2 - 5)}{(\sigma^2 + 1)^3(\sigma^2 + 2)^{5/2}} \right). \quad (11)$$

This shows that the optimum distance between the coils is equal to $2z_A = \sigma L = 0.544505643L$. Thus, the distance between the coils must be according to

$$2z_A = 0.544505643L. \quad (12)$$

With the optimum distance value, (9) can be rewritten in the form of (13). This describes the field generated by the system depending on the current, the number of turns

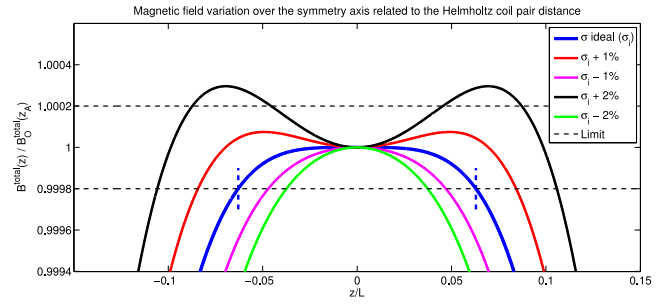


Fig. 3. Field variation for different distances between the coils used to determine the uniformity of the region according to the design requirements.

and the edge length of the square coil pair

$$|\mathbf{B}_O^{\text{total}}| = 1.628733676 \cdot 10^{-6} \frac{Ni}{L} \quad (13)$$

where the total field intensity, given in Tesla, is directly proportional to the number of turns N , to the current i and inversely proportional to the length L of the coil side.

C. Uniformity Region Analysis

Now that the ideal distance between the coils has been found, it is important to relate the edge (L) of the square coil to meet the 0.02% theoretical target decided as the factor of uniformity within the 11 cm^3 .

By means of the mathematical model shown before, it is possible to model the variation of the field related to σ [5]. Fig. 3 illustrates the percentage field variation over the shift in the axis of symmetry from the center of the coils, considering the optimum distance (σ_i), $\sigma_i \pm 1\%$ and $\sigma_i \pm 2\%$. This clearly shows the importance of keeping the correct distance between the windings of a coil pair.

Considering the ideal distance between a pair and assuming a uniformity of 0.02%, such theoretical limits will be given within $\pm 0.062956015L$ cm from the center. Consequently, for this project, the region that meets the desired uniformity will have a length of $0.12591203L$, where L is the coil size.

These constitute the final important theoretical aspects before deciding the coil size and its electrical parameters, which are presented in the Section IV.

IV. HELMHOLTZ COIL DESIGN

The magnetic field given by (13) is proportional to both the number of turns and the electrical current and is inversely proportional to the dimension of the coil. Therefore, both the structural and the electrical designs must be planned together. For example, any changes in the structural dimensions will implicate in a new group of values for parameters like number of turns, wire length, and uniformity within the cubic volume.

Also, some of the requirements presented in Section II are conflicting. For instance, the number of windings must be sufficient to the generate the field but small enough to avoid a high inductance. The same goes for the size of

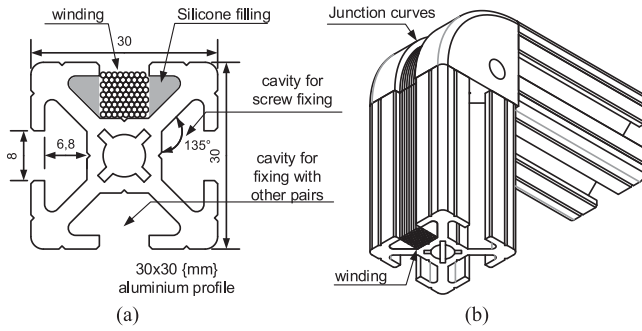


Fig. 4. Main structural profile used on the coils with the junction accessory to simplify wire winding. All dimensions are given in millimeters.

the structure, which must be only large enough to generate the uniformity volume, and above that will only result in smaller field production and higher assembly cost. The current itself should be as low as possible so that driving the system would not require more complex electronic circuits.

Therefore, the methodology to define the coil parameters was to first specify the structural design and the size of the coil, keeping it as low as possible while ensuring the required uniformity. Afterward, based on the structure, the wire choice was made considering aspects discussed in Section IV-B, but necessarily maintaining the electric current lower than 3 A and with a diameter appropriate for machine winding. Finally, the minimum number of windings required is computed and tested to be compatible with the desired structure.

A. Structural Design

Fig. 4(a) shows the chosen aluminum profile which is the main structural component of the coil. This profile was selected due to its mechanical strength, symmetry, and the presence of slots in its four external surfaces, which facilitates not only the winding procedure but also the fastening of other structural parts, such as plates and angle brackets.

A triaxial coil, such as the one described here, consists of the innermost, the middle, and the outermost pairs. These were designed with no gaps between each one and were mounted to produce a structure where the innermost coil must have the minimum principal dimension L as possible. Considering the requirement of an 11 cm region with a 0.02% field variation, the length L of the innermost winding is given as

$$L_{\text{Coil}}^{\text{inner}} = \frac{11.0}{0.12591203} \approx 88.0 \text{ cm} = 880 \text{ mm} \quad (14)$$

where $L_{\text{Coil}}^{\text{inner}}$ is the edge size of the innermost coil pair.

As there are no gaps between distinct pairs, the edge size for the middle, and the outermost ($L_{\text{Coil}}^{\text{middle}}$, $L_{\text{Coil}}^{\text{outer}}$) pairs can be found by adding twice the thickness of the structural aluminum profile employed. Since the profile has 30 mm of thickness, this results in 940 mm and 1000 mm, respectively.

Other auxiliary parts were manufactured to improve the mechanical strength and the assembly accuracy. The

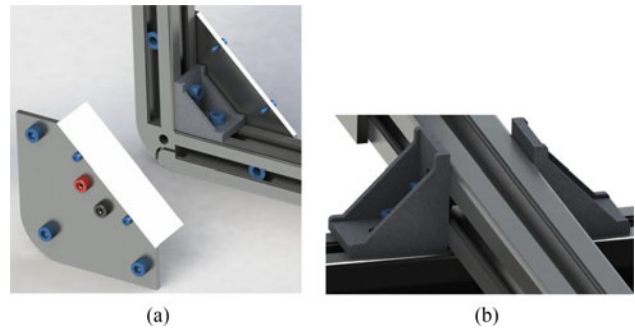


Fig. 5. (a) Corner detail of the assembly. (b) Fixing detail for two pairs of different sizes.

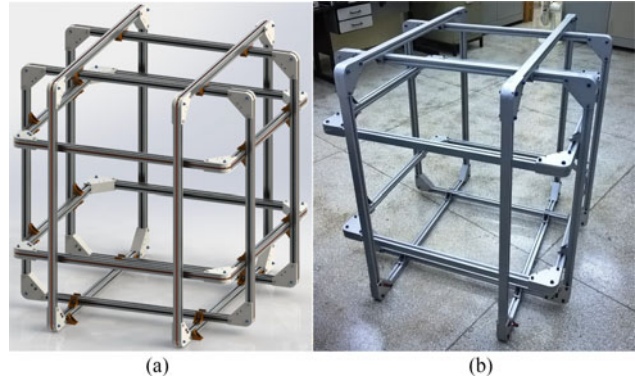


Fig. 6. (a) Triaxial Helmholtz CAD coil project, and (b) final assembly.

junction curve indicated in Fig. 4(b) was machined in aluminum, and its structural function is to connect the ends of the extruded profile bars that supports the winding. It also serves as a smooth rounded guide for the wire turns.

Fig. 5(a) shows the corner cover plate that holds two power input connectors (for banana jacks) for each winding and the internal structural fastening part which fixes two orthogonal bars. This last part is also used to fix the perpendicular windings of different coils, as shown in Fig. 5(b), producing the 3-D Helmholtz coil of Fig. (6).

B. Electrical Design

The electrical dimensioning of the project depends mainly on the wire gauge choice. To select a particular wire gauge, considerations were made regarding the ease of the coiling process, the maximum number of turns within the useful slot area of the profile, the resistance, and the maximum electrical current.

Wire gauges from 17 through 21 AWG best fit the requirements. Wires thicker than 17 AWG have a diameter inappropriate for winding and cannot be shaped into the coil's winding slot. Wires thinner than 21 AWG might not withstand the tension needed for coiling. Moreover, using thinner wires would limit the current, thus increasing the number of turns needed in each coil, which leads to higher inductance. Therefore, a wire gauge of 19 AWG was chosen. This choice is a compromise between simplifying the winding process and keeping a reduced number of turns per coil. Also, this wire supports currents levels of up to

TABLE I
Theoretical Electrical Parameters for the Design

Pair	N	Len. (m)	Res. (Ω)	Cur. (A)	Volt. (V)	Pwr (W)
Inn.	62	218.2	5.71	1.76	10.1	17.7
Mid.	66	248.2	6.49	1.76	11.4	20.1
Out.	70	280.0	7.32	1.76	12.9	22.7

1.76 A without overheating or losing its insulation capacity. As discussed below, with this current it is possible to guarantee a minimum field strength of 2 G using the setup built.

Once the maximum current and the required field strength have been determined, (13) is used to verify the number of turns needed in each of the coil pairs and also whether the number does not exceed $N_{\max}$ (mechanical limitation). Given the number of turns, it is possible to estimate the remaining electrical parameters, which are summarized in Table I. Note that the current, voltage, and power are maximum values while N is the minimum number of turns to obtain the required magnetic field intensity, considering that each pair of windings is connected in series. The maximum power and voltage values shown in Table I were calculated for a single axis coil. Hence, in order to drive 1.76 A in each pair, the power and voltage must be twice as high as shown.

C. Helmholtz Coil Prototype Results

Along with the steps above, a CAD project, illustrated in Fig. 6(a), was developed for validation. Fig. 6(b) shows the final assembled system for comparison. All the structural and fastening parts were included with the acquired aluminum profile. The junction curves and the edge auxiliary parts were machined. Though they were not 3-D-printed, the authors emphasize that acrylonitrile butadiene styrene printing would meet needs as well as the machined part at lower cost. To avoid magnetic field interference, all bolts and nuts used are made of aluminum.

The coil winding process was manual, but machine aided. Table II shows all the parameters measured and calculated with respect to the built prototype. The parameters were as expected by the design. A slightly greater number of turns was possible and recommended, since the number on the design was the minimum to obtain the necessary field without exceeding the maximum current. The resistance and inductance measured are almost the same for each axis of each pair, which shows that the winding process was accurate.

The final assembly and alignment of the system was done manually, aided by a mechanical measuring tape, a bubble level, and a square mark. A more accurate option would be the design of alignment parts to be used as assembly templates.

V. UNIFORMITY TEST METHODOLOGY

The test described in this section was performed to analyze the uniformity of the Helmholtz coil prototype. The

methodology was to measure the magnetic field using a fluxgate sensor FGMG3D [15], moving it along two axes, in order to evaluate the uniformity over an area around the center of the system. Fig. 7 shows the constructed wood table with slot guides that allows placing the sensor at points spaced every 0.5 cm on the z -axis and 3.5 cm on x -axis.

The measurements were performed using the setup described by the diagram of Fig. 8. Three Agilent E3634A power supplies configured as current sources were used to power the coils directly, in addition to two U1252A multimeters and one precision (8.5 digit) Agilent 3458A multimeter.

In order to perform the test, the FGM3D/250 sensor was initially placed exactly on the center of the Helmholtz coil. The current sources were adjusted so that the readings on all multimeters were as close as possible to 0 V, meaning that the geomagnetic field was canceled in all coordinates by the produced field. Next, a 2 A current was applied on the internal pair (y coordinate), creating a maximum field around it. The precision multimeter was connected to the y coordinate of the FGM3D sensor's reading. Measurements were obtained for the center slot of the guiding table, as well as for five more slots on each side. For each slot, 53 acquisitions were made using the 0.5 cm step marks, which is equivalent to a ± 26 cm spread.

It is important to consider that ferromagnetic materials, as well as any material that could interfere with the magnetic field, were placed as far as possible from the coil. Further discussion of interference issues can be found in the results.

VI. UNIFORMITY TEST RESULTS

Fig. 9 shows the measurements obtained by the 3458A multimeter considering all 53 points for all five slots closest to the center. A first look clearly shows that the highest value is not located in the center of either the x or the z coordinates. During the tests, it was observed that even with all the equipments turned OFF, the geomagnetic field over the table was not constant. This happens due to the presence of many materials that interfere with the geomagnetic field. In this particular case, the major contribution is due to an iron bar that serves as structure for the granite table where the coil structure was placed.

Therefore, even placing those parts as far as possible, the environment where the Helmholtz coil is installed in still causes considerable interference, both on the coils' generated field as well as on the geomagnetic one. Even a chair moving a meter away from the system would cause a field variation that could be captured by the 3458A multimeter.

Apart from that, the results of Fig. 9 exhibit the expected behavior. It is also notable that around ± 6 cm of the center, the measurements have minimum variation. To compare the practical uniformity with the theoretical requisite, such parameter was computed for a ± 5.5 cm around the center in each slot of Fig. 9. The worst slot measured 0.4% of uniformity in this 11 cm displacement. This measured uniformity fails to meet the project requirement by a factor of 2. The

TABLE II
Electrical Parameters Obtained on the Coil Prototype

Coil Type	N	Aprox. Coil Length (m)	Measured Resistance (Ω)		Max Current (A)	Max Voltage (V)	Max Power (W)	Aprox. Max. Field (G)	Inductance (mH)	
			A	B					A	B
Inner	63	225.2	5.89	5.89	1.76	10.4	18.3	2.052	8.9	11.1
Medium	72	272.7	7.13	7.13	1.76	12.6	22.1	2.150	15.0	13.6
External	73	291.0	7.60	7.61	1.76	13.4	23.5	2.093	15.8	16.3

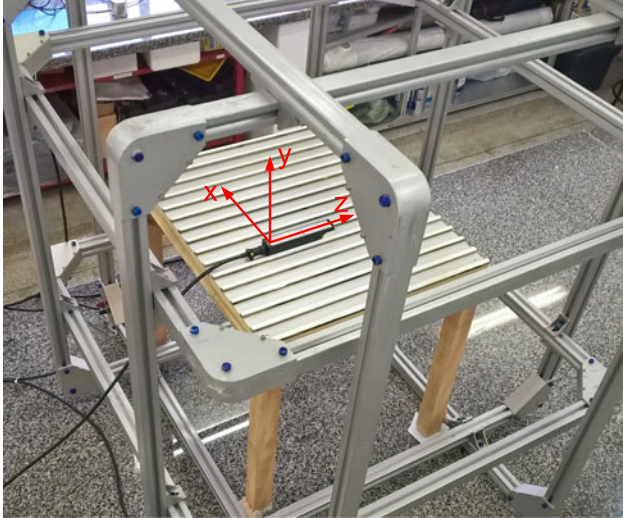


Fig. 7. Table used for alignment and displacement sensor over z - and x -axis

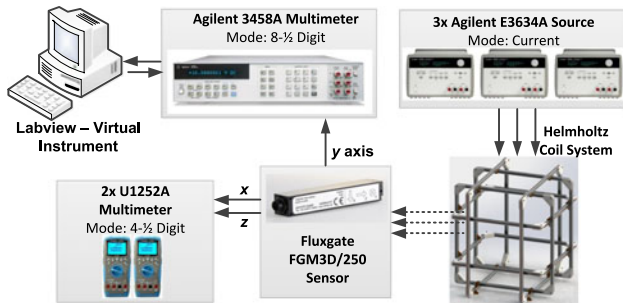


Fig. 8. Diagram of the setup used for the uniformity validation tests.

results for all measurements are presented in a mesh plot on Fig. 10.

Such a difference between the obtained and theoretical uniformity can be attributed to two factors: the interference caused by the ferromagnetic material present on the room and the manual assembly of the triaxial Helmholtz pair. To overcome these, the assembly must be improved to ensure the calculated distances between each axis' coils are correct. Also, the Helmholtz coil should be hosted in an environment without ferromagnetic elements.

For a better analysis of the uniformity, Table III is presented. For three distinct areas (7×7 , 14×14 , and 21×21 cm), three measurements are shown: the field strength at the origin, as well as the highest and lowest

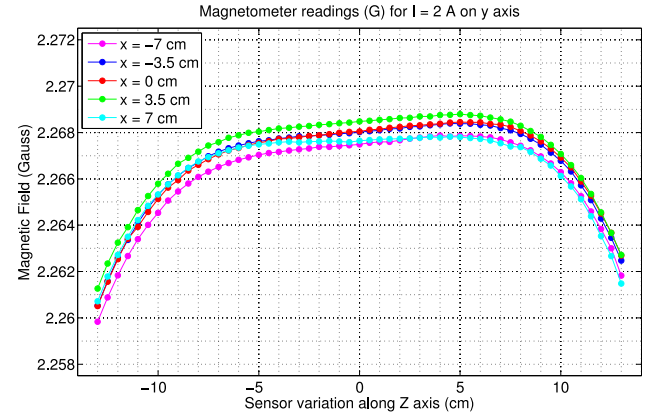


Fig. 9. Measures over the z - and x -axis for a constant current applied on the y -axis.

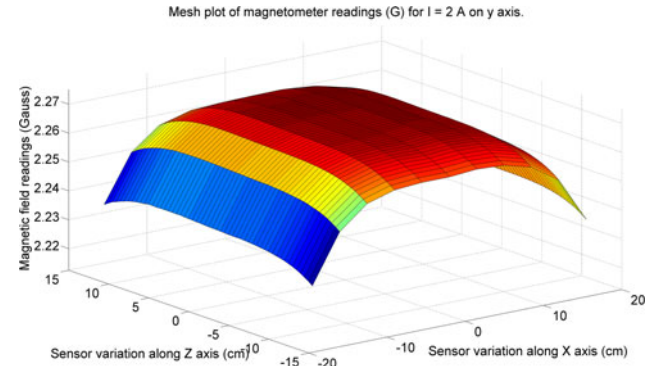


Fig. 10. Mesh plot of measured magnetic field for a constant current applied only on a single axis.

TABLE III
Results for Field Variation Within Different Areas

Area (cm)	Central Field (G)	Lowest Field (G)	Highest Field (G)	Field Var. (mG)	Var. (%)
7×7	2.268046	2.267746	2.268695	0.9488	0.0418
14×14	2.268046	2.266520	2.268790	2.2705	0.1001
21×21	2.268046	2.261249	2.268790	7.5417	0.3324

field strengths measured within the area. Field variation is also calculated and displayed in mG and percentage values. Since the sensor's step variation over the x coordinate

is 3.5 cm, the symmetrical square area analysis is only possible for 7, 14, and 21 cm dimensions.

Knowing that the presence and influence of ferromagnetic materials was inevitable over the generated field, an uniformity of 0.04% and 0.10% over an area of 7 cm × 7 cm and 14 cm × 14 cm, respectively, was considered a satisfactory result, given that it is similar to the technical specification for commercial Helmholtz coil systems and projects on the literature, such as [3], [7], [10], [16].

VII. CLOSED-LOOP TESTS SETUP

This section presents details of the closed-loop system used for driving the Helmholtz coil. For this first prototype, the chosen alternative was to implement a closed-loop system using a PID controller that would drive an independent voltage-controlled current source for each coil pair.

While the theory of PID controllers is relatively simple, implementing it either in an analog system using operational amplifiers or digitally using an embedded system can be a complex task, as well as tuning the PID parameters. Moreover, the current source must be symmetric and capable of supplying ± 2 A through a resistance of 14.2 Ω for the largest coil, which also requires an electronic system.

The current source chosen for implementation uses linear regulators and is based on the HCP. It will be addressed in Section VII-A. But considering that source, the most viable solution found for implementing a closed-loop system was to use a hardware-in-the-loop simulation solution compatible with MATLAB/Simulink tools. For this reason, modular simulation hardware from dSPACE¹ was selected. All dSPACE HiL hardware have their own MATLAB/Simulink library making it fully compatible and programmable using the Mathworks system. It provides the capability to simulate dynamic systems in the real time, while simultaneously injecting, reading, monitoring, and controlling analog and digital signals to the equipment under test in order to simulate its hardware and software.

The dSPACE hardware used is based on the DS1006 processor system and the DS2211 peripheral board. In essence, the MATLAB/Simulink model runs on the DS1006 processor whereas the dSPACE library provides access for all DS2211 peripherals. The fluxgate sensor is sampled by an analog-to-digital converter operating at 14-bit resolution, while the current source is controlled by a digital-to-analog converter operating at 12-bit resolution. Fig. 11 is a simplified block diagram of the system. Note that gray blocks are running digitally on the DS1006 processor.

The operating principle of the system is very simple. The difference between the setpoint and the measured value feeds the PID controller, whose output controls the plant. Next, the digital to analog converter (DAC) equivalent voltage needed for the field calculated by the controller is ob-

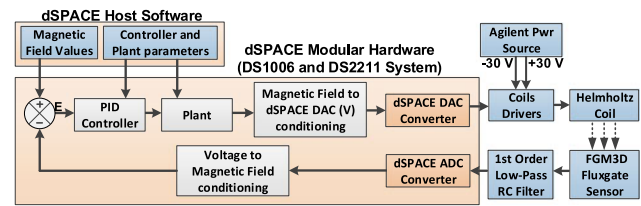


Fig. 11. System block diagram for using the Helmholtz coil prototype in a closed-loop system using a dSPACE hardware-in-the-loop simulation.

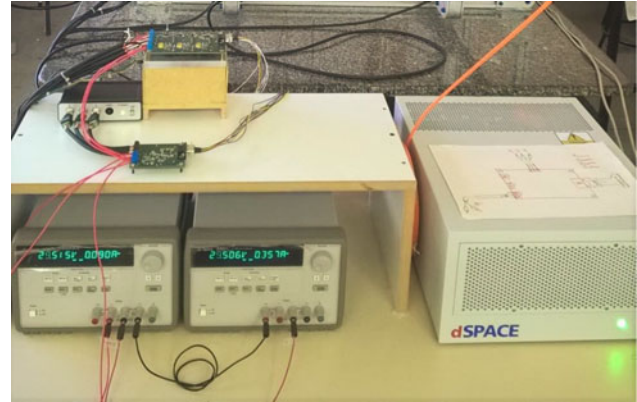


Fig. 12. Electronic setup for running the model described by Fig. 11.

tained and provided to the current source electronics. Signal conditioning is performed on the FGM3D sensor readings prior to connection with the analog to digital converter. Fluxgate acquisitions and the model are executed at 2 KHz, while the maximum update rate for the magnetic field input values is 100 Hz. Fig. 12 shows the actual setup at the laboratory. As can be seen, driving and controlling the system requires almost no electronics, due to the HiL simulation technique.

Furthermore, the earth's magnetic field is automatically compensated by the closed-loop controller. In practical terms, the geomagnetic measurements are simple offsets that are compensated by the controller.

A. Current Source

All discussions in this section refer to powering a single coil pair. As the project requires electrical currents up to 1.76 A, and assuming that the outermost pair has the highest resistance (14.2 Ω), powering each channel requires less than 30 W and 30 V. For this reason, it is feasible to implement current sources based on linear power operational amplifiers, which are simpler and less noisy than other alternatives.

Fig. 13 shows the schematic for the electronic system developed to supply the coils. The system has two stages: first, the DAC voltage (0–10 V) is conditioned to a symmetric value, which is input to the current drive, the second stage. The topology chosen for the current drive is based on the well-known HCP (HCP) [13].

This work implemented the standard HCP with slight improvements. The transfer function (TF) ($\frac{I_{out}}{V_{in}}$) is not trivial

¹This paper does not describe details of any dSPACE hardware used. Further information can be easily found in: <http://www.dspace.com/en/inc/home.cfm>.

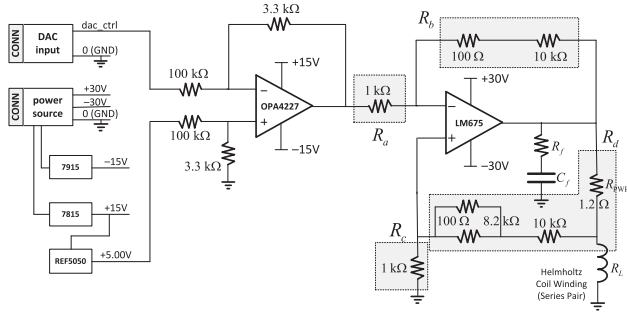


Fig. 13. Simplified schematic for one channel for the implemented current driver controlled by dSPACE DAC module based on the Howland current pump.

to be obtained and its deduction is not presented in here. Mahnam *et al.* [13] contains further analysis.

What is essential about the HCP is that the negative and the positive feedback impedance must be balanced. Considering the grouping of resistors presented in Fig. 13, if the relationship $R_a/R_b = R_c/R_d$ is met, the TF relating the load current to the voltage input is given as

$$\frac{I_{out}}{V_{in}} = -\frac{1}{R_{pwr}} \frac{R_b}{R_a}, \quad \text{if} \quad \frac{R_a}{R_b} = \frac{R_c}{R_d} \quad (15)$$

where I_{out} is the current through the power resistor (R_{pwr}) and V_{in} the signal applied to resistor R_a . Also, as the feedback current is negligible, the load current is given by I_{out} .

The power operational amplifier employed is the LM675 from Texas Instruments. This amplifier must have a minimum gain of 10 in closed-loop operation, which explains the choice of resistor values on the schematic. With that, it is possible to obtain the TF ($\frac{I_{out}}{V_{in}}$) to determine the symmetric signal necessary as input to the LM675. This value is obtained by the conditioning stage, which uses a precise 5V reference, obtained by a REF5050 integrated circuit from ACME, and a precision operational amplifier, OPA4427 from Texas Instruments, as a voltage subtractor circuit. This first stage is a basic circuit and would not be explained. The symmetric ± 30 V was supplied by two Agilent E3634A power supplies.

VIII. CLOSED-LOOP TESTS RESULTS

Section VIII-A presents the PID controller results. The open-loop step response for the coil is discussed and used to confirm the system TF and then used to design the controller. The simulation and practical results of the closed-loop step response are then introduced.

Section VIII-B presents actual results for the system described previously. Results considering different inputs are analyzed, but in all cases the closed-loop error was almost independent from the input.

A. Plant Parameters and PID controller

The system dynamics are dictated by the coil's resistor-inductor (RL) circuit in series with the power resistor (1.2Ω) of the HCP driver. The remaining elements are gains that convert the desired magnetic field to the

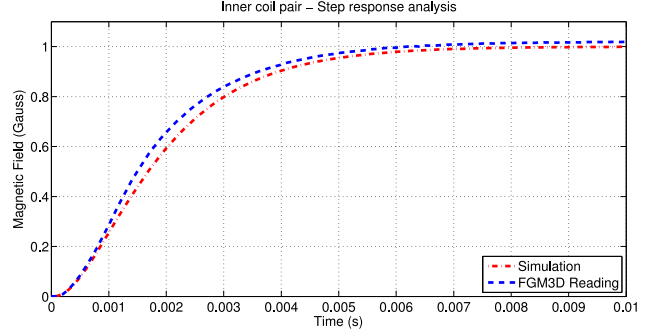


Fig. 14. MATLAB simulated step response for open-loop system and step response observed by dSPACE acquisition of the FGM3D sensor.

equivalent voltage needed by the current driver. Therefore, the plant's TF is that of an RL circuit, taking the resistance to be the sum of the resistive part of the coil and 1.2Ω . For the magnetic field measurement part, the TF is a series combination of the sensor's TF and the low-pass RC filter seen in Fig. 11. Since the filter cut-off frequency (120 Hz) is much lower than the sensor bandwidth (2 kHz), a dominant pole approximation to the low-pass filter TF can be employed.

The step response observed on the magnetic fluxgate sensor was simulated using MATLAB. The plant's TF for each pair can be directly obtained by the prototype electrical parameters of Table II. Moreover, results will be only shown for the innermost pair, since the analysis is equivalent for the remaining pairs. The open-loop response was also measured with the FGM3D sensor on the real system using the dSPACE setup. Fig. 14 shows both results.

Once the simulation results approximately matched the practical system, the digital PID controller design to run on dSPACE hardware was determined using MATLAB's *SISO Design tool* [17]. Since the controller study was not the main point in this work, the use of this tool to aid the PID development was a safe and fast way to obtain the digital controller parameters for the HiL simulation.

Considering that the dSPACE hardware runs the digital model at 2 kHz (equivalently, a $500 \mu s$ period), the controller topology which presented the best results (fast transient response with limited overshoot) was the MIGO technique (*M—constrained integral gain optimization*) [18]. The controller's TF for the innermost pair is given by (16), whereas Fig. 15 shows the step response simulation obtained by MATLAB as well as the measured response observed on the FGM3D sensor for the closed-loop model running on the dSPACE system

$$C(z) = \frac{3.3759z^2 - 3.7510z + 0.8752}{z^2 - z}. \quad (16)$$

The controller produced good results and the system behaved as expected for all pairs. It is important to note that the coil does not support continuous currents over 2.00 A. So, as a precaution, the PID output is saturated at the field strength value equivalent to this current. This limitation

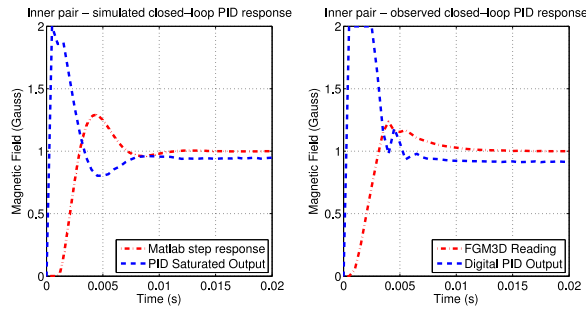


Fig. 15. MATLAB simulated step response for digital PID controller system and step response observed by dSPACE acquisition of the FGM3D sensor.

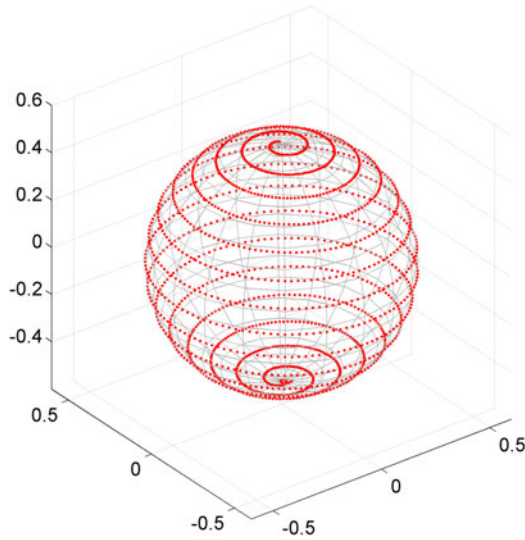


Fig. 16. Representation over the sphere of the generated magnetic field.

causes the system to have a longer transient period but does not affect the steady-state results.

At last, though the frequency response of the system is not shown, the closed-loop digital system has gain and phase stability for inputs of up to 50 Hz. As this coil's main applications are geomagnetic field simulation and sensor calibration, such frequency is suitable for the desired applications.

B. Magnetic Field Results

The wave selected to describe the closed-loop results is the representation of a body rotating over its axis subject to a constant magnetic field strength. Fig. 16 shows the expected points represented over the 3-D sphere. The generated field has a constant value of 500 mG. The field strength over time, for each individual axis, is described in Fig. 17.

Comparison shows that the field setpoints to the control system almost exactly match the FGM3D measurements in each axis. That is the reason why Fig. 17 actually shows the FGM3D sensor results. The difference (error) between the setpoint values and the field measurements by the fluxgate sensor is presented in Fig. 18.

The maximum error across the complete range is within $\pm 500 \mu\text{G}$. This value is actually better than the field

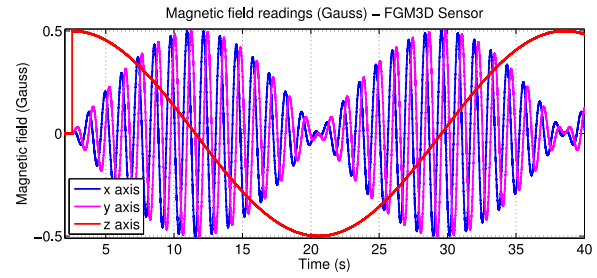


Fig. 17. FGM3D sensor readings using the closed-loop dSPACE HiL setup for the magnetic field points in Fig. 16 points.

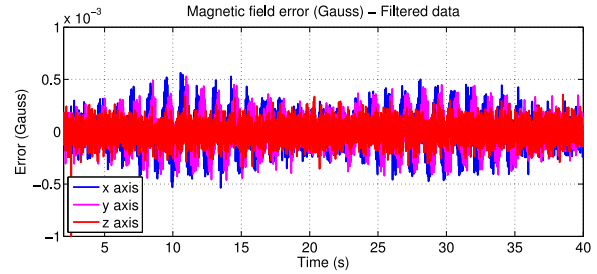


Fig. 18. Magnetic field error considering nonfiltered and filtered FGM3D readings using dSPACE HiL setup.

TABLE IV
Closed-Loop Results

	Inner Coil (y-axis)		Center Coil (x-axis)		Outer Coil (z-axis)		Module Analysis	
Error	P-P	RMSE	P-P	RMSE	P-P	RMSE	P-P	RMS
Value (μG)	994	158	1101	168	1070	171	1450	195

resolution achieved considering the dSPACE DS2211 DAC (12 bits) used to control the current driver for this coil prototype. Error results are given in tabular form in Table IV. The peak-to-peak (P-P) and the root-mean-square error (RMSE) were independently calculated for each axis and considering the field strength.

The result described above indicates that the system has enough frequency response to be used as a magnetic field simulator for the required applications. For example, it may be used to simulate the tumbling of a nanosatellite in orbit.

Considering applications to evaluate and test low-cost magnetic sensors, the RMSE error shows that the system built can perform well. Another important consideration is that fields with a different strength (e.g., $\pm 1000 \text{ mG}$) would show P-P and RMSE errors of the same order of magnitude. The hardware-in-the-loop simulation also reduces the complexity of the PID controller and makes it possible to run applications in parallel with it on the dSPACE hardware, keeping synchronization with the digital control loop.

IX. CONCLUSION

The triaxial Helmholtz coil design and implementation shown in this paper achieved great results considering the

whole system cost when compared to commercial magnetic field generation devices or similar coils found in the literature. As the use of machined or other complex parts was reduced with the use of simple commercial aluminum profiles and its accessories, the mechanical structure can be manufactured and assembled quickly and simply at a reasonable cost.

As the system will be mainly used for validation and tests of attitude determination and control systems that use magnetometers, the capability of generating magnetic fields up to 2 G is more than sufficient to cancel the earth's magnetic field and generate fields over 1.2 G in any axis. Moreover, the results show that it is possible to obtain a field uniformity of 0.04% and 0.10% within a 7 cm and a 14 cm sized cube around the coil's center, respectively. This volume can be considered ideal since nowadays magnetic devices such as commercial microelectromechanical systems sensors or even magnetic navigation equipment can be built with a very small footprint.

Another advantage of the system is the current needed to drive the coils. Since the maximum value is 1.76 A per axis, and considering each pair's resistance, less than 100 W is required to power all channels simultaneously. Therefore, it was possible to design a voltage-controlled current driver based on the HCP. The HCP presented and validated is simple, requiring only a few commercially available components and a single linear power regulator per axis.

Also, it is emphasized that the use of dSPACE modular hardware to perform a HiL simulation brings many benefits for the usage of the coil. The digital control implementation, in this case a PID controller, becomes simple and needs almost no electronics apart from the current driver itself.

The main drawback faced was the amount of interference due to ferromagnetic materials on the Helmholtz coil environment. Even so, the results presented were as satisfactory as those obtained from other coils described in the literature, while employing a simpler constructive design and a simpler electronic system.

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